ADHARSH SHOKKALINGAM

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EDUCATION

University of Alberta **Expected Graduation: May 2027**
Bachelor of Science, Computer Science, Minor in Mathematics *Edmonton, AB*

- **Relevant Coursework:** Software Engineering, Data Structures and Algorithms, Database Management, Computer Architecture I, II, Machine Learning, Operating Systems, Artificial Intelligence, Calculus I, II, III, IV

WORK EXPERIENCE

AlbertaSAT **Jun 2026 – Present**
Software Engineer *Edmonton, AB*

- Developing software for the on-board computer for the university **EX-ALTA3** mission using **Rust**, targeting bare-metal embedded systems with strict mission-critical reliability constraints
- Implemented Digital Fluxgate Magnetometer (**DFGM**) telemetry display formatting and currently building a confirmation safeguard for potentially dangerous spacecraft commands from the ground station

PROJECTS

3D Engine CPU Rasterizer | *C, SDL2*

- Engineered a fully interactive 3D engine in **C**, leveraging **SDL2** for graphics abstractions and input handling, with backface culling, near-plane clipping, dynamic camera transformations, and basic lighting
- Designed multiple real-time rendering modes (Wireframe, Vertex-Highlighted, Solid-fill, Textured) with keyboard-driven mode switching and **6** degrees of camera freedom
- Architected the project into modular components (Lighting, Geometry, Mesh, Culling, Camera, etc.), enabling extensibility and low-level control over the render pipeline

Sulfur-Events Lottery Application | *Java, Firebase*

- Developed a full-stack Android application with **4+** group members using **Java**, **Android Studio**, and **Firebase** for managing and tracking event/user **data**
- Implemented **QR** code generation and scanning features, including storing and extracting embedded **geolocation** data to display and update event/user locations
- Utilized **Firebase Cloud Firestore** for structured cloud data storage, **querying**, and multi-user data consistency

RISC-V to WASM Converter | *RISC-V (RV32I)*

- Designed and implemented support for R-Type, I-Type, and Branch instructions, including arithmetic, logic, shifts, and conditional control flow
- Built a signed **LEB128** encoder from scratch to extract immediate operands and convert them to WASM bytecode

Distributed System Monitor | *C++*

- Built a distributed monitoring system across two programs, distpsnotify and remoteagent, coordinating **real-time data collection** over **TCP/IP** sockets from multiple remote machines simultaneously
- Designed and implemented a custom client-server command protocol (**EXECUTE/QUIT**, **START/STOP** delimiters) for structured, ordered data exchange across persistent socket connections
- Engineered dynamic **memory management** for an arbitrary number of remote agents using heap allocations, verifying **zero** memory leaks across all code paths using **Clang** leak tools on macOS

Consistent | *Swift, Supabase (AI – Agentic Development)*

- Native **iOS** habit-tracking application (**SwiftUI**, **Supabase**/**Postgres** with Row-Level Security, offline-first sync) built using an **AI agent**-driven engineering workflow directing code generation and system architecture specifications

OS 1 | *C, QEMU* **In Progress**

- Building a bare-metal **x86_64 OS** kernel with a custom **IDT** for **CPU** exceptions and hardware interrupt handling, multilevel paging for virtual memory, and a heap allocator supporting bump, linked-list, and fixed-size block strategies

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, Rust, RISC-V, HTML/CSS, SPSS
Libraries/Frameworks: PostgreSQL, MongoDB, Firebase, Supabase, Spring Boot, UML, Valgrind, Clang, Makefile, CMake, QEMU
Developer Tools: Git, MongoDB, Docker, Android Studio